

Comparison

1. DB_BENCH Results

Benchmark	Category	Description	Threads	Entries	µs/op	ops/sec	Seconds	Total Ops	MB/s	Notes
fillseq	Write-Only	Sequential fill	1	10M	1.932	517,654	19.3	10M	57.3	WAL enabled
fillseqdeterministic	Write-Only	Deterministic seq fill	1	10M	2.894	345,564	31.8	11M	38.2	Auto-compact off, 3 levels
fillrandom	Write-Only	Random fill (16 threads)	16	10M	364.738	43,866	3647.5	160M	4.9	Very slow (compaction)
filluniquerandomdeterministic	Write-Only	Unique random fill	1	10M	4.765	209,877	52.4	11M	23.2	Auto-compact off, 3 levels
fillsync	Write-Only	Synchronous fill	1	1M	4.380	228,322	4.4	1M	25.3	sync=true
fill100K	Write-Only	Large values (100KB)	1	1M	120.583	8,293	120.6	1M	791.0	Large values
overwrite	Write-Only	Overwrite existing	16	10M	367.352	43,553	3673.6	160M	4.8	use_existing_db=1
deleteseq	Write-Only	Sequential delete	1	1M	1.847	541,293	1.8	1M	-	use_existing_db=1
deleterandom	Write-Only	Random delete	1	1M	4.463	224,067	4.5	1M	-	use_existing_db=1
readseq	Read-Only	Sequential read	16	10M	1.482	9,839,494	14.6	144M	1088.5	FASTEST
readreverse	Read-Only	Reverse seq read	16	10M	1.840	8,203,435	17.6	144M	907.5	use_existing_db=1
readrandom	Read-Only	Random read	16	10M	39.620	400,045	400.0	160M	39.8	89.98% found
readmissing	Read-Only	Read non-existent	16	1M	59.360	263,791	60.7	16M	-	0% found (expected)
readtocache	Read-Only	Cache warm-up	1	10M	0.476	2,102,206	4.3	9M	232.6	use_existing_db=1
multireadrandom	Read-Only	Batched multi-read	4	5M	9.542	407,817	49.0	20M	36.1	batch_size=8
readwhilewriting	Mixed R-W	Read under write	16	10M	62.877	254,299	60.1	15.3M	25.4	write_rate_limit=2MB/s
readwhilemerging	Mixed R-W	Read under merge	16	10M	80.900	196,345	814.9	160M	20.7	merge_operator=put
readrandomwriterandom	Mixed R-W	Mixed R+W	16	10M	-	CRASHED	-	-	-	Segmentation fault
updaterandom	Mixed R-W	Read-modify-write	16	10M	886.195	18,052	8863.0	160M	4.0	write_rate_limit=2MB/s
seekrandomwhilewriting	Mixed R-W	Scan under write	16	10M	140.035	114,014	1403.3	160M	141.8	seek_nexts=10

2. YCSB Results

GC = Garbage Collection - Java Virtual Machine internal operations.

Workload	Description	Record Count	Threads	Run Time (ms)	Throughput (ops/s)	GC Count	GC Time (ms)	GC %
A (Load)	Insert 1M records	1M	8	9,574	104,449.55	63	128	1.337
A (Run)	50% Read, 50% Update	1M	8	13,465	74,266.62	57	139	1.032
B	95% Read, 5% Update	1M	8	6,746	148,235.99	30	77	1.141
C	100% Read	1M	8	6,419	155,787.51	27	65	1.013

Workload	Description	Record Count	Threads	Run Time (ms)	Throughput (ops/s)	GC Count	GC Time (ms)	GC %
D	Read Latest + Insert	1M	8	2,175	459,770.11	23	60	2.759
E	Short Ranges (Scan)	1M	8	41,002	24,389.05	415	1,324	3.229
F	Read-Modify-Write	1M	8	9,174	109,003.71	59	122	1.330

3. YCSB Detailed Latency Breakdown (µs)

Workload	Operation	Count	Avg	Min	P50	P95	P99	Max
A (Load)	INSERT	1,000,000	73.20	4	33	276	575	172,671
A (Run)	READ	499,724	53.97	1	37	121	348	10,719
A (Run)	UPDATE	500,276	140.21	7	67	493	1,224	170,111
B	READ	949,684	46.21	1	36	97	237	11,655
B	UPDATE	50,316	83.16	8	61	153	533	30,239
C	READ	1,000,000	46.05	1	36	97	205	10,695
D	READ	949,665	13.18	1	4	37	69	15,311
D	INSERT	50,335	40.46	5	19	72	385	46,783
E	SCAN	949,873	335.20	6	299	621	999	21,071
E	INSERT	50,127	55.13	8	38	103	265	53,791
F	READ	1,000,000	37.55	1	31	79	175	9,191
F	UPDATE	500,051	59.02	5	28	144	510	152,191

4. Cross-Tool Comparison (Equivalent Workloads)

Workload Pattern	db_bench Benchmark	db_bench ops/s	YCSB Workload	YCSB ops/s	Good
Mixed R-W (50/50)	readrandomwriterandom	CRASHED	Workload A	74,267	YCSB (only option)
Read-Modify-Write	updaterandom	18,052	Workload F	109,004	YCSB (6x faster)
Read-Heavy (95/5)	readwhilewriting*	254,299	Workload B	148,236	db_bench (1.7x)
100% Read	readrandom	400,045	Workload C	155,788	db_bench (2.6x)
Scan + Write	seekrandomwhilewriting	114,014	Workload E	24,389	db_bench (4.7x)
Sequential Write	fillseq	517,654	Workload A Load	104,450	db_bench (5x)

Performance based on throughput

1. DB_BENCH (Fast to Slow)

Rank	Benchmark	ops/sec	Category
1	readseq	9,839,494	Read-Only
2	readreverse	8,203,435	Read-Only
3	readtocache	2,102,206	Read-Only
4	deleteseq	541,293	Write-Only
5	fillseq	517,654	Write-Only
6	multireadrandom	407,817	Read-Only
7	readrandom	400,045	Read-Only
8	fillseqdeterministic	345,564	Write-Only
9	readmissing	263,791	Read-Only
10	readwhilewriting	254,299	Mixed R-W
11	fillsync	228,322	Write-Only
12	deleterandom	224,067	Write-Only
13	filluniquerandomdeterministic	209,877	Write-Only
14	readwhilemerging	196,345	Mixed R-W
15	seekrandomwhilewriting	114,014	Mixed R-W
16	fillrandom	43,866	Write-Only
17	overwrite	43,553	Write-Only
18	updaterandom	18,052	Mixed R-W
19	fill100K	8,293	Write-Only
-	readrandomwriterandom	CRASHED	Mixed R-W

2. YCSB (Fast to Slow)

Rank	Workload	ops/sec	Description
1	D	459,770	Read Latest (in-memory)
2	C	155,788	100% Read
3	B	148,236	95% Read, 5% Update
4	F	109,004	Read-Modify-Write
5	A (Load)	104,450	Bulk Insert
6	A (Run)	74,267	50% Read, 50% Update
7	E	24,389	Short Ranges (Scan)

Observations

1. DB_BENCH and 1M YCSB

Sr. No.	Observation	Note
1	(Read) Sequential > Random	readseq (9.8M) is 24.6x faster than readrandom (400K)
2	Random writes are expensive	fillrandom (43K) is 11.8x slower than fillseq (517K)
3	Sync kills throughput	fillsync is 2.3x slower than fillseq
4	Compaction stalls	Max latency spikes to 172ms+ in YCSB (compaction I/O)
5	Scan is slowest YCSB	Workload E at 24K ops/s (335 µs/scan)
6	Latest reads are fastest	Workload D at 459K ops/s (13 µs/read) all in-memory
7	db_bench is more aggressive	16 threads, raw C++ vs YCSB's 8 threads, Java JNI
8	Critical bug found	db_bench readrandomwriterandom segfaults in RocksDB 11.6.0
9	GC matters for scans	Workload E: 3.2% GC time vs 1.0% for reads
10	Tail latency is high	P99 << Max everywhere - typical of LSM stores under compaction

2. 1M YCSB

Workload	Throughput	Read Latency	Write Latency	Characteristics
A (50/50)	74,267 ops/sec	53.97 µs	140.21 µs	Balanced R/W; update cost 2.6x higher
B (95% R)	148,236 ops/sec	46.21 µs	83.16 µs	Read-heavy; 2x throughput vs A
C (100% R)	155,788 ops/sec	46.05 µs	N/A	Pure read; best throughput
D (Latest)	459,770 ops/sec	13.18 µs	40.46 µs	Cache-friendly; 6x throughput vs A
E (Scan)	24,389 ops/sec	N/A	55.13 µs	Scan heaviest - 81% slower than C
F (RMW)	109,004 ops/sec	37.55 µs	59.02 µs	RMW adds overhead vs pure read

3. 10M YCSB

Workload	Throughput	Read Latency	Write Latency	Scaling vs 1M
A (10M, 50/50)	99,618 ops/sec	69.86 µs	783.80 µs	Write latency 5.6x ↑
B (10M, 95% R)	127,343 ops/sec	81.00 µs	N/A	Read latency 1.75x ↑

4. Scaling Impact (1M → 10M)

Workload	1M Throughput	10M Throughput	Change	Read Lat 1M	Read Lat 10M	Lat Change
A (50/50)	74,267	99,618	+34%	53.97 μs	69.86 μs	+1.3x
B (95% R)	148,236	127,343	-14%	46.21 μs	81.00 μs	+1.75x
C (100% R)	155,788	134,508	-14%	46.05 μs	98.90 μs	+2.15x
D (Latest)	459,770	184,788	-60%	13.18 μs	11.68 μs	-0.88x
E (Scan)	24,389	24,389	0%	N/A	N/A	Same benchmark